AN IMPROVED UPPER BOUND FOR THE UNIVALENT BLOCH–LANDAU CONSTANT

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Abstract. We exhibit a disc with 24 slits, 18 of them curved, and give a computer-assisted proof that its ratio of inradius to conformal radius is less than 0.65628589. This is below the best previously proved upper bound 0.6563937 for the univalent Bloch–Landau constant (Carroll and Ortega-Cerdà, 2009) and below the best numerical value 0.656319277 in the literature (Ortega-Cerdà and Pridhnani, 2010). In the proof, the inradius is bounded by an exact covering argument, and the conformal radius is bounded below by a comparison function whose boundary values are controlled in ball arithmetic. The domain was found by an evolutionary search driven by a spectral integral-equation solver.

1. Introduction

Let $\mathbb{D}$ be the unit disc. For a domain $D \subset \mathbb{C}$ write $R(D)$ for its *inradius*, the supremum of the radii of the discs contained in D. The *univalent Bloch–Landau constant*, also called the univalent or schlicht Bloch constant, is

$$\mathcal{U} = \inf\big\{R(f(\mathbb{D})) :\ f \text{ univalent in } \mathbb{D},\ |f'(0)| = 1\big\};$$

thus $f(\mathbb{D})$ contains discs of every radius less than $\mathcal{U}|f'(0)|$ for every univalent f on $\mathbb{D}$. If $D \neq \mathbb{C}$ is simply connected and $0 \in D$, write $\rho(D)$ for its *conformal radius* at 0, that is $|f'(0)|$ for a conformal map f of $\mathbb{D}$ onto D with $f(0) = 0$. Applying the definition to $f/f'(0)$,

$$\mathcal{U} \le \frac{R(D)}{\rho(D)} \qquad \text{for every simply connected } D \neq \mathbb{C} \text{ with } 0 \in D, \tag{1}$$

and $\mathcal{U}$ is the infimum of the right-hand side. The bounds in Table 1 were all obtained from (1) with D a disc from which slits have been removed.

	upper bound	domain: a disc minus…
Robinson [8] (1935)	0.658	radial slits
Goodman [4] (1945)	0.65647	radial slits in bisecting generations
Beller, Hummel [1] (1985)	0.6564155	radial slits, third generation not bisecting
Carroll, Ortega-Cerdà [3] (2009)	0.6563937	six radial slits, six harmonically symmetric arcs
Ortega-Cerdà, Pridhnani [6] (2010)	0.656319277	as [3], with more such arcs

Table 1. Upper bounds for $\mathcal{U}$. The value of [3] is the numerical evaluation of an explicit formula. The values of [1] and [6] are numerical; the authors of [6] state that they have not done a rigorous error analysis. The best lower bound known to us is 0.5708858 [9].

Theorem 1. $\mathcal{U} < 0.65628589$.

The domain D that proves Theorem 1 is described in Section 2. The proof has two independent parts:

$$R(D) \le 1 + 10^{-9} \quad \text{(Proposition 3.1)}, \qquad \rho(D) \ge 1.523726208329 \quad \text{(Proposition 4.1)},$$

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and $(1+10^{-9})/1.523726208329 < 0.65628589$. Both parts rely on computations, carried out in exact integer arithmetic and in ball arithmetic with the Arb library [5]; the programs are part of the repository accompanying this paper and run in about one minute. Section 5 describes how the domain was found and the independent numerical checks, none of which is used in the proof.

The domains of [3, 6] are built from arcs that are harmonically symmetric, a condition which is necessary for the boundary arcs of an extremal domain that do not touch an extremal disc [2] and which makes those domains amenable to explicit computation. No such condition is imposed here: the slits are taken from a four-parameter family and optimised numerically. The resulting domain nevertheless resembles the domain of [6] (Figure 1).

2. The domain

Fix $R > 0$. A *slit* with parameters (θ, t, b_1, b_2), $0 < t < R$, is the arc

$$z = \exp w(x), \qquad w(x) = L + ax + i\big(\theta + b_1(ax)^2 + b_2(ax)^4\big), \qquad -1 \le x \le 0, \tag{2}$$

where $L = \log R$ and $a = \log(R/t) > 0$. Since $|z| = Re^{ax}$, the slit starts on the circle $|z| = R$ at $Re^{i\theta}$, its modulus decreases monotonically along it, and it ends at a tip of modulus t. For $b_1 = b_2 = 0$ it is the radial segment $[te^{i\theta}, Re^{i\theta}]$.

Let $R = 5.001117628756668$. The slits of our domain are those with the parameters (θ, t, b_1, b_2) of Table 2 together with their images under the group generated by $z \mapsto e^{2\pi i/3}z$ and $z \mapsto \bar{z}$, that is, the slits with parameters $(2\pi r/3+\varepsilon\theta,\ t,\ \varepsilon b_1,\ \varepsilon b_2)$, $r = 0, 1, 2$, $\varepsilon = \pm 1$. For classes 0 and 1, where $b_1 = b_2 = 0$ and $\theta \in \{0, \pi/3\}$, the two signs give the same three slits; for the other classes they give six. This gives 24 slits in five classes. All decimal numbers in this section are to be read as exact rational numbers.

class	θ	t	b_1	b_2	slits
0	0	1	0	0	3
1	$\pi/3$	1.9999999974092333	0	0	3
2	0.029930188727091613	2.789342409403121	0.9026921410641155	−0.3301459494660587	6
3	1.0157094972537686	3.473381248030611	−1.2656124700253997	0.9794269477754911	6
4	0.5757776442359243	4.342874184977691	0.03225521024585182	2.9360081049550737	6

Table 2. The parameters of the domain; $R = 5.001117628756668$.

Let K be the union of the 24 slits and let D be the connected component of $\{|z| < R\} \setminus K$ that contains 0.

Lemma 2.1. *D is a simply connected domain, $0 \in D \neq \mathbb{C}$, and $\partial D \subset \{|z| = R\} \cup K$. In particular* (1) *applies to D.*

Proof. Every point of K has modulus at least 1, so $0 \notin K$. Each slit is connected and meets the circle $|z| = R$, so $E = \{|z| \ge R\} \cup \{\infty\} \cup K$ is a closed connected subset of the Riemann sphere. Every component of the complement of a closed connected subset of the sphere is simply connected, and D is such a component. The statement about the boundary is clear. □

Remark 2.2. Numerically, the slits are pairwise disjoint (the smallest distance between two of them is about 0.1484), so that $D = \{|z| < R\} \setminus K$; this is not needed for the proof.

3. The inradius

Proposition 3.1. *$R(D) \le 1 + 10^{-9}$.*

The proof rests on two elementary observations.

Lemma 3.2. *If the disc of centre p and radius r is contained in D and $q \notin D$, then $r \le |p-q|$.*

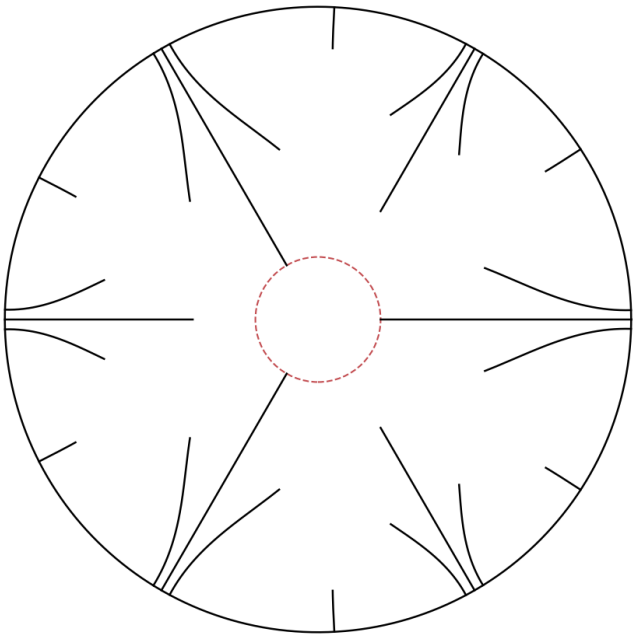


FIGURE 1. The domain D. The disc $|z| < 1$ is one of 22 discs of radius between $1 - 4 \times 10^{-6}$ and 1 that are locally largest.

Lemma 3.3. *Let C be a closed square, $q \in \mathbb{C}$ and $\lambda > 0$. If $|p - q| \le \lambda$ at the four corners p of C, then $|p - q| \le \lambda$ for all $p \in C$.*

Proof. $p \mapsto |p - q|$ is convex, and C is the convex hull of its corners. □

Proof of Proposition 3.1. Put $\lambda = 1 + 10^{-9}$. The domain is invariant under $z \mapsto e^{2\pi i/3} z$ and $z \mapsto \bar{z}$, so it suffices to show that no disc of radius greater than λ contained in D has its centre in $S = \{p : \ 0 \le \arg p \le \pi/3, \ |p| < R\} \cup \{0\}$. Note $S \subset [0, 8]^2$.

The program `inradius_certificate.py` produces a finite family of closed squares (*cells*) of the form $[i2^{3-\ell}, (i+1)2^{3-\ell}] \times [j2^{3-\ell}, (j+1)2^{3-\ell}]$ with integers $\ell \ge 6$ and $0 \le i, j < 2^\ell$, and verifies the following.

(i) *The cells cover* $[0, 8]^2$. The cells are pairwise distinct, none is contained in another, and their areas add up to 64; both statements are checked in exact arithmetic. Since two dyadic squares are either nested or have disjoint interiors, the cells tile the square.

(ii) Each cell $[x_0, x_1] \times [y_0, y_1]$ is of one of three kinds:
   (a) $x_0^2 + y_0^2 \ge R^2$. Then the cell does not meet the open disc $|p| < R$.
   (b) $y_0^2 > 3x_1^2$. Then every point of the cell has $\arg p > \pi/3$, so the cell does not meet S.
   (c) A point $q \notin D$ is specified with $|p - q| \le \lambda$ at the four corners of the cell. By Lemmas 3.3 and 3.2, no disc of radius greater than λ contained in D has its centre in the cell.

In (c) the point q is a point $\exp w(x)$ of one of the slits (2), with $x \in [-1, 0]$, or a point of the circle $|z| = R$; such points do not belong to D. For a slit point the parameter is given as a double-precision number $s = ax$ (an exact rational), and $-a < s \le 0$ is verified in ball arithmetic, or it is the tip $s = -a$. The coordinates of q are enclosed in ball arithmetic and replaced by a dyadic rational point q' with $|q - q'| < 2^{-69}$; the inequality actually checked, in exact integer arithmetic, is $|p - q'|^2 \le (\lambda - 2^{-69})^2$ at the four corners. The inequalities in (a) and (b) involve only rational numbers and are checked exactly.

The run for the domain of Section 2 produces 4993 cells: 2799 of kind (a), 464 of kind (b) and 1730 of kind (c), the latter using 610 distinct points q. The smallest cells have side 2^{-29}. □

The cells and the points q are found by a floating-point search which plays no role in the proof: only its output is verified.

4. The conformal radius

Proposition 4.1. *$\rho(D) \geq 1.523726208329$.*

4.1. The comparison principle.

Lemma 4.2. *Let v be continuous on $\overline{D} \setminus \{0\}$, harmonic in $D \setminus \{0\}$, with $v(z) = -\log|z| + c + o(1)$ as $z \to 0$ and $v \leq 0$ on ∂D. Then $\log \rho(D) \geq c$.*

Proof. Let φ map D conformally onto $\mathbb{D}$ with $\varphi(0) = 0$, and $g = -\log|\varphi|$. Then $g > 0$ is harmonic in $D \setminus \{0\}$ and $g(z) = -\log|z| + \log \rho(D) + o(1)$ at 0, because $\varphi(z) = z/(\rho(D)e^{i\gamma}) + O(z^2)$. The function $g - v$ is harmonic in $D \setminus \{0\}$ and bounded near 0, hence harmonic in D. At every boundary point $\liminf(g - v) \geq -\limsup v \geq 0$, since $g > 0$ and v is continuous with $v \leq 0$ on ∂D. By the minimum principle $g - v \geq 0$ in D, and its value at 0 is $\log \rho(D) - c$. □

4.2. The comparison function.

We work in the variable $w = \log z$ and use the notation of (2), with a subscript j for the j-th slit. We extend w_j to $-1 \leq x \leq 1$ by the same formula and call $\Gamma_j = w_j([-1, 1])$ an *arc*; the slit is $w_j([-1, 0])$ and the rest, $w_j((0, 1])$, lies in the half-plane $\operatorname{Re} w > L$. The reflection $\sigma(w) = 2L - \bar{w}$ in the line $\operatorname{Re} w = L$ satisfies

$$\sigma(w_j(x)) = w_j(-x), \tag{3}$$

because the imaginary part of w_j is even in x. Put

$$\Phi(w) = -\log\bigl|2\sinh(w/2)\bigr|.$$

Φ is harmonic in $\mathbb{C} \setminus 2\pi i\mathbb{Z}$, it is $2\pi i$-periodic, $\Phi(-w) = \Phi(\bar{w}) = \Phi(w)$, and

$$\Phi(w) = -\tfrac{1}{2}|\operatorname{Re} w| - \log\bigl|1 - e^{-|\operatorname{Re} w| - i \operatorname{Im} w \operatorname{sgn}(\operatorname{Re} w)}\bigr| = -\tfrac{1}{2}|\operatorname{Re} w| + O(e^{-|\operatorname{Re} w|}) \quad (|\operatorname{Re} w| \to \infty), \tag{4}$$

where $\operatorname{sgn}(0) = 1$.

For each slit let ψ_j be a real polynomial that is *odd*, with Chebyshev expansion $\psi_j = \sum_{n \text{ odd}} c_{j,n} T_n$, and assume that $\psi_j = \psi_{j'}$ whenever the slits j and j' belong to the same class. Define

$$U(w) = (L - \operatorname{Re} w) + \sum_j \int_{-1}^{1} \Phi\bigl(w - w_j(y)\bigr)\, \psi_j(y)\, \frac{dy}{\sqrt{1 - y^2}}. \tag{5}$$

Lemma 4.3. *(a) U is continuous on $\mathbb{C}$, $2\pi i$-periodic, and harmonic outside $\bigcup_j (\Gamma_j + 2\pi i\mathbb{Z})$.*
(b) $U(\sigma(w)) = -U(w)$; in particular $U = 0$ on the line $\operatorname{Re} w = L$.
(c) $U(w) = -\operatorname{Re} w + c + o(1)$ as $\operatorname{Re} w \to -\infty$, uniformly in $\operatorname{Im} w$, where

$$c = L - \frac{\pi}{4} \sum_j a_j c_{j,1}. \tag{6}$$

(d) $U(w + 2\pi i/3) = U(w)$ and $U(\bar{w}) = U(w)$.

Consequently $u(z) = U(\log z)$ is well defined and continuous on $\{0 < |z| \leq R\}$, harmonic in $\{0 < |z| < R\} \setminus K$, zero on $|z| = R$, invariant under the symmetries of D, and $u(z) = -\log|z| + c + o(1)$ as $z \to 0$.

Proof. (a) Periodicity and harmonicity follow from the properties of Φ. For continuity, substitute $y = \cos\vartheta$: each term of (5) is $\int_0^\pi \Phi(w - w_j(\cos\vartheta))\psi_j(\cos\vartheta)\, d\vartheta$, the potential, with kernel Φ, of a measure ν with bounded density with respect to $d\vartheta$. Since $\operatorname{Re} w_j(\cos\vartheta) = L + a_j \cos\vartheta$ and the set $\{\vartheta \in [0, \pi] : |\cos\vartheta - \xi| < \eta\}$ has length at most $C\sqrt{\eta}$ for all ξ, every disc of radius η has $|\nu|$-measure at most $C'\sqrt{\eta}$. Splitting the integral over $\{|w - \zeta| < \eta\}$ into

the annuli $2^{-k-1}\eta \le |w-\zeta| < 2^{-k}\eta$ gives $\sup_w \int_{|w-\zeta|<\eta} \bigl|\log|w-\zeta|\bigr|\, d|\nu|(\zeta) \to 0$ as $\eta \to 0$. As $\Phi(w) + \log|w|$ is continuous near 0, the potential is therefore the uniform limit, as $\eta \to 0$, of the potentials with the continuous kernels $\min(\Phi, \log(1/\eta))$, and so it is continuous. (For background on logarithmic potentials see [7].)

(b) $L - \operatorname{Re} w$ changes sign under σ. Since $\sigma(w) - \sigma(w') = -\overline{(w-w')}$ and $\Phi(-\bar w) = \Phi(w)$, (3) gives $\Phi(\sigma(w) - w_j(y)) = \Phi(w - w_j(-y))$; substituting $y \to -y$ and using that ψ_j is odd shows that each integral changes sign.

(c) By (4), for $\operatorname{Re} w \to -\infty$ and w' in a bounded set, $\Phi(w-w') = \frac12(\operatorname{Re} w - \operatorname{Re} w') + o(1)$ uniformly. Now $\int_{-1}^{1} \psi_j \frac{dy}{\sqrt{1-y^2}} = 0$ because ψ_j is odd, and $\operatorname{Re} w_j(y) = L + a_j y$ with $\int_{-1}^{1} y\,\psi_j(y) \frac{dy}{\sqrt{1-y^2}} = \frac{\pi}{2} c_{j,1}$ by the orthogonality of the Chebyshev polynomials. Hence the j-th integral tends to $-\frac{\pi}{4} a_j c_{j,1}$.

(d) The slit with parameters $(\theta + 2\pi/3, t, b_1, b_2)$ is the translate by $2\pi i/3$ of the slit with parameters (θ, t, b_1, b_2), and the slit with parameters $(-\theta, t, -b_1, -b_2)$ is its complex conjugate, with the same parametrisation by x. The family of arcs is invariant under both operations modulo $2\pi i$, the polynomials ψ_j depend only on the class, and $\Phi(\bar w) = \Phi(w)$.

The consequences for u follow, noting that the parts $w_j((0,1])$ of the arcs lie in $\operatorname{Re} w > L$. □

Proposition 4.4. *Let $\delta \ge 0$ be such that $u \le \delta$ on K. Then $\log \rho(D) \ge c - \delta$, with c given by (6).*

Proof. Apply Lemma 4.2 to $v = u - \delta$: by Lemma 2.1, $\partial D \subset \{|z| = R\} \cup K$, and $v = -\delta \le 0$ on the circle, $v \le 0$ on K. □

Proposition 4.4 holds for arbitrary odd polynomials ψ_j. In the proof of Proposition 4.1 they are obtained from a floating-point solution of the integral equation $U = 0$ on the arcs (Section 5): the coefficients are averaged over each class, the even-indexed ones are set to zero, and the result, in double precision, is stored in a data file. The quality of that solution affects only the size of δ.

4.3. **Bounding u on a slit.** By Lemma 4.3(d) it is enough to consider one slit i in each class. Let

$$f_i(x) = U(w_i(x)), \qquad -1 \le x \le 1.$$

By Lemma 4.3(b) and (3), f_i is odd, so $\sup_K u \le \max_i \sup_{[-1,1]} |f_i|$. We write $\|\cdot\|$ for the supremum norm on $[-1,1]$, $d\mu = dy/\sqrt{1-y^2}$, and use two kernels:

$$G_{ij}(x,y) = 2\sinh\frac{w_i(x) - w_j(y)}{2} \quad (j \ne i), \qquad K_i(x,y) = \frac{2\sinh\frac{w_i(x)-w_i(y)}{2}}{x-y},$$

the latter extended analytically across $x = y$. Explicitly, $K_i = \Delta_i \operatorname{shc}\bigl(\frac12(x-y)\Delta_i\bigr)$ with $\operatorname{shc}(t) = \sinh(t)/t$ and the divided difference $\Delta_i(x,y) = a_i + i\bigl(b_{1,i}a_i^2(x+y) + b_{2,i}a_i^4(x^3 + x^2y + xy^2 + y^3)\bigr)$. For real x, y,

$$\begin{aligned} \Phi\bigl(w_i(x) - w_j(y)\bigr) &= -\log|G_{ij}(x,y)| \qquad (j \ne i), \\ \Phi\bigl(w_i(x) - w_i(y)\bigr) &= -\log|x-y| - \log|K_i(x,y)|. \end{aligned}$$

Lemma 4.5. *For $n \ge 1$ and $-1 \le x \le 1$,* $\displaystyle\int_{-1}^{1} -\log|x-y|\, T_n(y)\, d\mu(y) = \frac{\pi}{n} T_n(x).$

Proof. With $x = \cos\phi$, $y = \cos\vartheta$ one has $-\log|\cos\phi - \cos\vartheta| = \log 2 + \sum_{m\ge1} \frac{2}{m} \cos m\phi \cos m\vartheta$ in L^2, and $\int_0^\pi \cos m\vartheta \cos n\vartheta\, d\vartheta = \frac{\pi}{2}\delta_{mn}$. □

Consequently, for real x,

$$f_i(x) = -a_i x + \sum_n c_{i,n}\frac{\pi}{n}T_n(x) - \int_{-1}^{1}\log|K_i(x,y)|\,\psi_i(y)\,d\mu(y) - \sum_{j\neq i}\int_{-1}^{1}\log|G_{ij}(x,y)|\,\psi_j(y)\,d\mu(y). \tag{7}$$

For $r>1$ let E_r be the closed region bounded by the ellipse with foci ±1 and semi-axes $\frac12(r\pm r^{-1})$. For $b>0$ let $Q_b=[-\sqrt{1+b^2},\sqrt{1+b^2}]\times[-b,b]$; then $E_r\subset Q_b$ when $r=b+\sqrt{1+b^2}$, since the semi-axes of E_r are $\sqrt{1+b^2}$ and b.

Lemma 4.6 (Chebyshev coefficients from the real part)**.** *Let h be analytic in a neighbourhood of E_r, and $|\operatorname{Re}h|\le A$ on ∂E_r. Then the real function $\operatorname{Re}h|_{[-1,1]}$ has a uniformly convergent Chebyshev expansion $\alpha_0+\sum_{k\ge1}\alpha_kT_k$ with*

$$|\alpha_k|\le\frac{4A}{r^k+r^{-k}}\le 4Ar^{-k}\qquad(k\ge1).$$

Proof. The function $H(\zeta)=h(\frac12(\zeta+\zeta^{-1}))$ is analytic in a neighbourhood of the annulus $r^{-1}\le|\zeta|\le r$ and $H(\zeta)=H(1/\zeta)$, so its Laurent coefficients satisfy $\beta_{-k}=\beta_k$. On $|\zeta|=1$ this gives $h(\cos\vartheta)=\beta_0+\sum_{k\ge1}2\beta_k\cos k\vartheta$, hence $\alpha_k=2\operatorname{Re}\beta_k$ for $k\ge1$. On $|\zeta|=r$, whose image is ∂E_r,

$$\frac{1}{2\pi}\int_0^{2\pi}\operatorname{Re}H(re^{i\vartheta})e^{-ik\vartheta}\,d\vartheta=\tfrac12\big(\beta_kr^k+\overline{\beta_{-k}}\,r^{-k}\big)=\tfrac12\big(\beta_kr^k+\overline{\beta_k}\,r^{-k}\big).$$

The left-hand side has modulus at most A, and the real part of the right-hand side is $\frac12\operatorname{Re}\beta_k\,(r^k+r^{-k})$. □

Note that h need not be real on $[-1,1]$, and that only a bound for $\operatorname{Re}h$ is required.

Lemma 4.7 (Gauss–Chebyshev quadrature)**.** *Let F be a real continuous function on $[-1,1]$ whose Chebyshev coefficients satisfy $|\alpha_k|\le Cr^{-k}$ for $k\ge1$, with $r>1$, and let ψ be a polynomial of degree d. Let $y_l=\cos\frac{(2l-1)\pi}{2N}$, $l=1,\dots,N$, and $\kappa=2N-1-d\ge0$. Then*

$$\Big|\int_{-1}^{1}F\psi\,d\mu-\frac{\pi}{N}\sum_{l=1}^{N}F(y_l)\psi(y_l)\Big|\le\frac{2\pi C\|\psi\|}{r-1}\,r^{-\kappa}.$$

Proof. Write $F=P+\varrho$ with P the Chebyshev series of F truncated at degree κ; then $\|\varrho\|\le\sum_{k>\kappa}Cr^{-k}=Cr^{-\kappa}/(r-1)$. The rule is exact for polynomials of degree at most $2N-1$, hence for $P\psi$, and both the integral and the rule applied to $\varrho\psi$ are bounded by $\pi\|\varrho\|\|\psi\|$. □

Lemma 4.8 (Interpolation)**.** *Let f be continuous on $[-1,1]$ with Chebyshev coefficients $|\phi_k|\le Cr^{-k}$ for $k\ge1$, $r>1$, and let $p=\sum_{q<M}\gamma_qT_q$ be the polynomial of degree less than M that interpolates f at the zeros of T_M. Then*

$$\|f\|\le\sum_{q<M}|\gamma_q|+\frac{2C}{r-1}\,r^{1-M}.$$

Proof. At the zeros $x_l=\cos\frac{(2l+1)\pi}{2M}$ of T_M and for $k=2Mm\pm k'$ with $0\le k'\le M$ one has $T_k(x_l)=(-1)^mT_{k'}(x_l)$, and $T_M(x_l)=0$. So the interpolant of T_k, $k\ge M$, is $\pm T_{k'}$ with $k'<M$, or 0, and $\|T_k-\text{interpolant}\|\le2$. Since interpolation is linear and reproduces polynomials of degree less than M, $\|f-p\|\le2\sum_{k\ge M}|\phi_k|\le2Cr^{1-M}/(r-1)$. Finally $\|p\|\le\sum|\gamma_q|$. □

4.4. **The certificate.** Fix a class representative i. The program `conformal_certificate.py` performs the following steps. All bounds below (the kernel bounds, $\|\psi_j\|$, A_{ij}, B_i, the remainders, the values at the nodes and the final result) are computed in ball arithmetic. Floating-point numbers are used only to choose the parameters b, b'_{ij}, M, N_{ij} and the boxes of the branch and bound, where a poor choice could only make a step fail, and to pass enclosures between processes as a double-precision midpoint with an upward-rounded radius. The program also verifies that the even-indexed coefficients in the data file are zero, and that each ellipse lies in its rectangle.

Step 1: kernel bounds in the variable x. A number $b > 0$ is chosen and, for every j, numbers $0 < \lambda_{ij} \le \Lambda_{ij}$ are computed with

$$\lambda_{ij} \le |G_{ij}(x,y)| \le \Lambda_{ij} \quad (j \ne i), \qquad \lambda_{ii} \le |K_i(x,y)| \le \Lambda_{ii} \qquad \text{for } x \in Q_b,\ y \in [-1,1]. \tag{8}$$

This is done by branch and bound: the set $Q_b \times [-1,1]$, of real dimension three, is bisected into boxes until on each box the ball-arithmetic evaluation of the kernel gives an enclosure excluding 0, with upper bound less than four times the lower bound. Put $A_{ij} = \max(|\log \lambda_{ij}|, |\log \Lambda_{ij}|)$.

Since the kernels are continuous and do not vanish on the simply connected set $Q_b \times [-1,1]$ (nor on a neighbourhood of it), they have continuous logarithms there, analytic in x. Thus

$$\begin{aligned} h_i(x) = & -a_i x + \sum_n c_{i,n} \frac{\pi}{n} T_n(x) - \int_{-1}^{1} \log K_i(x,y)\, \psi_i(y)\, d\mu(y) \\ & - \sum_{j \ne i} \int_{-1}^{1} \log G_{ij}(x,y)\, \psi_j(y)\, d\mu(y) \end{aligned}$$

is analytic in a neighbourhood of $Q_b \supset E_{r_i}$, where r_i is slightly less than $b + \sqrt{1+b^2}$, and $\operatorname{Re} h_i = f_i$ on $[-1,1]$ by (7). Write $r = r_i$. On E_r we have $|\operatorname{Re} x| \le \frac{1}{2}(r + r^{-1})$, $|T_n(x)| \le \frac{1}{2}(r^n + r^{-n})$ and $|\operatorname{Re} \log(\cdot)| = |\log |\cdot|| \le A_{ij}$, so

$$|\operatorname{Re} h_i| \le B_i := \tfrac{1}{2} a_i (r + r^{-1}) + \sum_n |c_{i,n}| \frac{\pi}{2n} (r^n + r^{-n}) + \pi \sum_j \|\psi_j\| A_{ij} \quad \text{on } E_r,$$

where $\|\psi_j\|$ is bounded by $\sum_n |c_{j,n}|$. By Lemma 4.6 the Chebyshev coefficients of f_i satisfy $|\phi_k| \le 4 B_i r_i^{-k}$.

Step 2: kernel bounds in the variable y. For every j a number $b'_{ij} > 0$ is chosen and bounds of the form (8) are computed in the same way for $x \in [-1,0]$ (real) and $y \in Q_{b'_{ij}}$, giving A'_{ij}. For each fixed $x \in [-1,0]$, $y \mapsto \log G_{ij}(x,y)$ (respectively $\log K_i(x,y)$) is then analytic near E_r, $r = b'_{ij} + \sqrt{1 + b'^2_{ij}}$, with real part bounded by A'_{ij}, and Lemmas 4.6 and 4.7 show that the N-point Gauss–Chebyshev rule computes the corresponding integral in (7) with an error of at most

$$\frac{8\pi A'_{ij} \|\psi_j\|}{r-1}\, r^{-(2N-1-d)}, \qquad d = \deg \psi_j. \tag{9}$$

$N = N_{ij}$ is chosen so that this is at most 10^{-13}.

Step 3: values at the nodes. M (even) is chosen so that $8 B_i r_i^{1-M}/(r_i - 1) \le 10^{-9}$. At each zero $x_k < 0$ of T_M an enclosure of $f_i(x_k)$ is computed from (7): the first two terms directly, each integral by the Gauss–Chebyshev rule with N_{ij} nodes, adding the bound (9) to the radius. The values at the nodes $x_k > 0$ follow from the oddness of f_i.

Step 4: conclusion. The coefficients γ_q of the interpolant are computed from the enclosures of the $f_i(x_k)$ by the discrete cosine transform, in ball arithmetic. By Lemma 4.8,

$$\sup_{[-1,1]} |f_i| \le \delta_i := \sum_{q<M} |\gamma_q| + \frac{8 B_i}{r_i - 1} r_i^{1-M}.$$

Then $\delta = \max_i \delta_i$ satisfies the hypothesis of Proposition 4.4, and the program evaluates $\exp(c - \delta)$ in ball arithmetic, with c from (6), and rounds it down.

4.5. **Result.**

Proof of Proposition 4.1. The coefficients $c_{j,n}$, $n \le 63$, are stored in the repository, in the data file `density_G_n64.json`. With them, the program gives the values in Table 3, hence $\delta = 2.656 \times 10^{-8}$, $c = 0.42115881436\ldots$ and $\rho(D) \ge \exp(c - \delta) \ge 1.523726208329$. □

class	a_i	r_i	B_i	M	N_{ij}	δ_i
0	1.6097	1.00934	9.4	3196	79–2225	2.66×10^{-8}
1	0.9165	1.01733	6.6	1676	78–1118	3.55×10^{-9}
2	0.5839	1.00805	8.7	3716	78–2088	1.76×10^{-8}
3	0.3645	1.01548	5.7	1872	78–1087	3.76×10^{-9}
4	0.1411	1.34403	4.1	88	76–153	7.48×10^{-10}

TABLE 3. Data of the certificate for the five classes. The small values of $r_i - 1$ for classes 0 and 2, and 1 and 3, reflect the proximity of the slits of these classes: about 0.03 in the w-plane.

Remark 4.9 (Validation). The same program, applied to the two slit discs whose conformal radius is known in closed form, returns lower bounds slightly below the exact values, as it must:

domain	exact ρ	certified lower bound
$\mathbb{D} \setminus [\frac{1}{2}, 1)$	$8/9 = 0.888888888889$	0.888888888327
six-slit disc (see below)	1.520743036313	1.520743034966

The six-slit disc is $\{|z| < 3\}$ minus the segments $[1,3)e^{2\pi ik/3}$ and $[2,3)e^{\pi i(2k+1)/3}$, $k = 0, 1, 2$. The first value is $\rho(\mathbb{D}\setminus[s,1)) = 4s/(1+s)^2$ with $s = \frac{1}{2}$, which follows from the Koebe function. For the second, with $R = 3$ and $t = 2$: let $s = R^{-3}$ and let g be the conformal map of $\mathbb{D} \setminus [s, 1)$ onto $\mathbb{D}$ with $g(0) = 0$, $g'(0) > 0$; it maps $(-1, 0)$ into itself. With $-\beta = g(-(t/R)^3)$, applying $z \mapsto z^3$, then g, then a cube root, then the same procedure for the slit $[\beta, 1)$, gives $\rho = R\left(4s/(1+s)^2\right)^{1/3}\left(4\beta/(1+\beta)^2\right)^{1/3}$.

Proof of Theorem 1. By Lemma 2.1 and (1), and Propositions 3.1 and 4.1,

$$\mathcal{U} \le \frac{R(D)}{\rho(D)} \le \frac{1 + 10^{-9}}{1.523726208329} = 0.65628588360\ldots < 0.65628589. \qquad \square$$

5. How the domain was found, and independent checks

Nothing in this section is used in the proofs.

The solver. The Green function of D is sought in the form (5), with ψ_j a polynomial of degree less than n. Requiring $U = 0$ at the n zeros of T_n on every arc, with the logarithmic part of the self-interaction integrated exactly (Lemma 4.5) and all other integrals computed by the n-point Gauss–Chebyshev rule, gives a linear system of size $24n$ with condition number of order 10^3. Then $\rho \approx \exp(c)$ with c from (6). In practice the method converges very fast: for D the value $\rho = 1.523726248805$ is the same for $n = 64$, 96 and 128. For slit discs of the two types of the Remark above it agrees with the closed-form values to rounding error (at most 3×10^{-16}).

The search. The parameters R, the tip modulus of class 1 and (θ, t, b_1, b_2) for classes 2 to 4 were optimised by differential evolution [10], followed by a simplex search, the objective being the ratio of a floating-point inradius (local maxima of the distance to the boundary, located on a grid and refined by Newton's method) to the computed conformal radius. As a check, the best disc with six radial slits gives 0.657793, in agreement with [8]; the best disc with twelve radial slits that we found gives 0.6564163, slightly above the value 0.6564155 of [1].

Independent recomputation. Separately written floating-point programs, using different methods, reproduced the values for D: a branch and bound over the whole disc found no disc of radius greater than $1 + 2 \times 10^{-8}$, and a boundary integral equation in the z-plane, with a parametrisation adapted to the tips and no use of symmetry, gave $\rho(D) = 1.5237262488052$, stable under changes of the discretisation and under the substitution $\zeta = z^3$.

The numerical value. The floating-point value of the ratio is 0.656285866; the proved bound is weaker by 1.8×10^{-8}.

6. Remarks

(1) *What is trusted.* The proof relies on the correctness of the Arb library [5] and its Python interface, of integer arithmetic in Python, and of the two certificate programs, which together comprise about 650 lines.
(2) *Optimality.* The domain is the output of a numerical search in a small family. We have not examined whether its arcs are harmonically symmetric, and there is no reason to expect the bound of Theorem 1 to be sharp.
(3) *Other domains.* A simpler domain with 18 slits, six radial and twelve curved with $b_2 = 0$, gives $\mathcal{U} < 0.65630853$ by the same programs.

Declaration on the use of AI

The domain, the numerical method, the programs and the proof in this paper were produced with the help of an AI system (Claude, Anthropic), under my direction throughout.

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